

RESEARCH ARTICLE

Multivariate distribution correction of climate model outputs: A generalization of quantile mapping approaches

Léonard Dekens^{1,2} | Sylvie Parey¹ | Mathilde Grandjacques³ | Didier Dacunha-Castelle⁴¹MFEE, EDF Recherche et Développement, Chatou, France²Ecole Normale Supérieure de Lyon et Université Claude Bernard Lyon 1, France³CEA, LIST, Laboratoire d'Intelligence Artificielle pour les Nouvelles Énergies, Le Bourget Le Lac, France⁴Université Paris Sud, Orsay, France**Correspondence**

Sylvie Parey, MFEE, EDF Recherche et Développement, 6 Quai Watier 78401, Chatou, France.

Email: sylvie.parey@edf.fr

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Climate change impact studies necessitate the estimation of climate variable evolution in the future. These are given by climate model simulations made under different greenhouse gas and aerosol emission scenarios agreed at the international level. However, climate model outputs have biases, especially at the local scale, and need to be corrected against observations. Common bias correction methods are distribution based and form the well-known quantile mapping approaches. This paper presents a generalization of such techniques to the consideration of multivariate distributions. This approach uses the basic lemma of Lévy and Rosenblatt, which allows the transport of a distribution on another one, in every dimension. It needs convenient nonparametric estimations of conditional repartitions. The approach is first tested in a controlled framework, by use of statistical simulations, then in the real setting of climate simulation, in the bivariate case. An important issue of these types of distribution corrections is the different kinds of hypotheses of stationarity over a long enough period: stationarity of the link between model and observations whatever the period or stationarity of the change between the present and future for model and observations. This choice differentiates approaches like quantile mapping and cumulative distribution function transform, for example, in the univariate framework, and makes them more efficient, in the univariate as well as in the multivariate context, when the data to be corrected best verify the assumed hypothesis.

KEYWORDS

bias adjustment, climate change, multivariate distributions, statistics

1 | INTRODUCTION

Because climate change is now attested (Intergovernmental Panel on Climate Change (IPCC), 2013), and mitigation still underway, adaptation has to be anticipated in parallel to mitigation. The first step in adaptation is an estimation of the possible consequences of climate change at the scale of human societies and their activities. These estimations are commonly done through impact studies, based for example on specific models run with climatic variables. Observed variables are used to represent current conditions, whereas climate model outputs are used to project future conditions. Climate models are numerical tools based on physical representations of the dynamics of the components, atmosphere, ocean, ice or land surface, and of their interactions, through physical or biochemical processes. Although such tools are more and more sophisticated, including more and more detailed processes, their outputs may still differ significantly from the local observations commonly used by impact models. Therefore, bias correction and downscaling techniques have become an active area of research in the last decade or so.

The approach here, like other quantile mapping approaches, can be used for bias adjustment or bias adjustment and downscaling depending on the spatial scale of the reference data set. When used with local observations, it aims at predicting, in statistical terms, local climate variables using more global data provided by a climate model working at a larger scale.

Statistical bias correction methods are widely used to correct the distribution of the climate model variables so that they match that of some local observations. Such techniques are commonly recommended for impact studies (Chen, Brissette, Chaumont, & Braun, 2013; Gudmundsson, Bremnes, Haugen, & Engen-Skaugen, 2012; Teutschbein & Seibert, 2012). The most used techniques are the so-called quantile mapping approaches (Déqué, 2007; Haddad & Rosenfeld, 1997; Panofsky & Brier, 1958; Piani, Haerter, & Coppola, 2010; Wood, Leung, Sridhar, & Lettenmaier, 2004) and their variants like the cumulative distribution function transform (CDFt; Michelangeli, Vrac, & Loukos, 2009). However, a limitation of such techniques is that the correction is applied independently to the different variables when more than one climatic variable is needed for an impact study, with the risk of degrading the consistency between them. Recent approaches have been proposed to tackle this caveat, and correct two variables, essentially temperature and rainfall, in a consistent way (Li, Sinha, Horton, Diffenbaugh, & Michalak, 2014; Piani & Haerter, 2012; Zhang & Georgakakos, 2012). Vrac and Friederichs (2015) go further and propose an approach, based on the empirical copula (by reordering univariate bias-corrected variables), potentially able to tackle both the intervariable and spatial consistencies. However, one issue with the approach is that it can only reproduce the historical temporal sequencing, which is an important limitation for climate impact studies. Cannon (2016) suggests a methodology based on the correction of the marginal distributions again by quantile mapping and then an iterative scheme to push either the Pearson correlation dependence structure or the Spearman rank correlation dependence structure toward observed values.

The approach proposed and tested in this paper is a generalization of the quantile mapping techniques to the correction of multivariate distributions. The chosen setting is the typical problem faced with impact studies: Over a historical period, time series of different climate variables are available from both climatic databases and climate model simulations, whereas for a future period, necessarily, only the climate model time series are available. Then, as climate model outputs have biases compared to the observations, the aim of the correction is to estimate, for the future period, specific characteristics of time series at the desired location closer to that of the observations. Then, what is expected is not the precise sequencing in time of the variables, which is not an expected result of climate models, but rather characteristics as their distribution, which are quite invariant for time periods with adequate length, that is, not too long to be able to neglect climate trends but not too short to be able to estimate characteristics such as the distributions. The methodology will be described in the fully multivariate context, considering p dimensions, but in practice, a dimension larger than 2 means much longer time series for the distribution estimations. This methodology uses, as a basic trick, the transportation of a distribution on $\mathbb{R}^p$ onto another one fixed in advance. This is done by repeated applications of the lemma of Lévy and Rosenblatt (Grandjacques, 2015; Grandjacques, Delinchant, & Adrot, 2015). This approach allows clarifying which kinds of stationarities are required and also gives a natural way of making clear which period lengths are concerned by all these approaches.

The theory underlying the methodology is presented in Section 2, and Section 3 explains the estimation choices made. Then, Section 4 presents an application in a controlled framework, by use of bivariate Gaussian distributions, in order to evaluate when our bivariate bias correction is best justified. Bivariate Gaussian distributions are easy to handle, although climate variables are generally not normally distributed. The aim here is only to better understand the involved transformations. Finally, Section 5 is devoted to an application to real climate variables, before coming to the conclusion and discussion in Section 6.

2 | THEORETICAL FRAMEWORK

2.1 | The problem to be solved

As previously exposed, the aim here is to simultaneously bias-correct different climatic variables according to available observations. We have then two time series with values in $\mathbb{R}^p$:

1. a time series $(Y_t)_{t \in P_0}$ given over the period $P_0 = (1, \dots, n_0)$ only and corresponding to the observations;
2. a time series (X_t) over a much larger period and given by a numerical climate model simulation for example.

The aim is then to obtain for a future period P , later than P_0 , a projection of some characteristics of $(Y_t)_{t \in P}$, based on assumptions made about the link between X and Y . In order to make it easier in the following, we will suppose that P and P_0 are of the same length and P corresponds to $P_k = \{kn_0 + 1, \dots, (k+1)n_0\}$.

The proposed methodology is based on the following assumptions.

Hypothesis H₁. For each $k \in \mathbb{N}$, there is a process $X_{k,t}$, restriction of X_t when $t \in P_k$, which is stationary and weakly mixing. Similarly, it is supposed that there is a process $(Y_t)_{t \in \mathbb{N}}$ stationary and weakly mixing, for which $(Y_t)_{t \in P_0}$ is a restriction to the period P_0 .

H₁ allows dealing with the intrinsic nonstationarity of the data by selecting long enough periods over which it can reasonably be considered as stationary. If n_0 is large enough, this assumption allows valid estimations for some characteristics such as the distributions of $(Y_t)_{t \in P_0}$ or $(X_{k,t})_{t \in P_k}$ for k in $\mathbb{N}$, because both processes have good ergodic properties. “Large enough” here means sufficiently large to apply the law of large number for the needed estimations but not too large to avoid too strong trend effects. The distributions of $(Y_t)_{t \in P_0}$ and $(X_{k,t})_{t \in P_k}$ are respectively denoted by F_0 and G_k .

Hypothesis H₂. F_0 and G_k are continuous with densities f_0 and g_k , respectively, with f_0 and g_k , $k \in \mathbb{N}$ strictly positive on the interior of their support.

H₂ avoids having to take conventions to compute inverse functions for the cumulative distributions functions, which we will have to consider in the methodology (in fact, there is of course no inverse repartition in dimension > 1 but a set of p one-dimensional ones, which are defined later using the Lévy–Rosenblatt lemma). Such conventions would lead to very complex formulations. If one of the components of Y_t is continuous with the exception of a mass on one point, as is the case for rainfall, what follows can easily be extended to consider such a behavior.

2.2 | Distribution transfer on $\mathbb{R}^p$

Definition. If F and G are distributions defined on $\mathbb{R}^p$ and verifying H_2 , and U^p the uniform distribution on $\mathbb{R}^p$, product of p uniform distributions on $\mathbb{R}$, then transferring distribution G to F will be defined as an application $T: \mathbb{R}^p \rightarrow \mathbb{R}^p$, such that $T(G) = F$, $T(G)$ being the distribution image of G through T .

In what follows, T is first defined as an application $T: \mathbb{R}^p \rightarrow \mathbb{R}^p$ and then used as operator on distributions, defined for any Borelian set I by $T(G)(I) = G(T_H^{-1}(I))$, H being the distribution of I .

The notations used for the conditional distributions and their inverses are the following:

If Z is a vector with distribution H on $\mathbb{R}^p$, $Z = (Z^1, Z^2, \dots, Z^p)$, F_1 is the distribution of Z^1 , and for each k in $\{2, \dots, p\}$, $F_{k/1, \dots, k-1}(z_1, \dots, z_k)$ is the conditional distribution of Z^k for fixed $Z^1, \dots, Z^{k-1}$. Thus,

$$F_{k/1, \dots, k-1}(z_1, \dots, z_{k-1}, z_k) = P(Z^k \leq z_k | Z^1 = z_1, \dots, Z^{k-1} = z_{k-1})$$

Then, the inverse function of each strictly increasing function $h: \mathbb{R} \rightarrow \mathbb{R}$ is denoted by h^{-1} .

The following lemma proven by Paul Lévy and better known as the Rosenblatt lemma (Grandjacques, 2015; Grandjacques et al., 2015) defines a transfer function for each distribution H (verifying H_2 in our case) on $\mathbb{R}^p$. H is used here for genericity and stands for any distribution as F or G mentioned earlier.

Lemma 1. If U is defined as

$$\begin{cases} U^1 = H_1(Z^1) \\ U^k = H_{k/1, \dots, k-1}(Z^1, \dots, Z^k) \\ U^p = H_{p/1, \dots, p-1}(Z^1, \dots, Z^p), \end{cases}$$

then the distribution of $U = (U^k)_{k=1, \dots, p}$ is U^p , uniform with independent marginal distributions.

Thus, if T_H is the previously described transformation, $T_H(H) = U^p$ and its inverse $T_H^{-1}: \mathbb{R}^p \rightarrow \mathbb{R}^p$, such that $T_H^{-1}(U^p) = H$ is defined as

$$\begin{cases} Z^1 = H_1^{-1}(U^1) \\ Z^k = H_{k/1, \dots, k-1}^{-1}(U^1, \dots, U^k) \\ Z^p = H_{p/1, \dots, p-1}^{-1}(U^1, \dots, U^p). \end{cases}$$

Remark. T_H is obviously not the only transformation allowing a distribution transfer; different versions can be proposed depending, for example, on the order according to which each component of Z is considered. T_H is sequential with respect to the space dimension; this is a useful property in the applications. Note that, in the Gaussian case, it is not at all the classical transformation obtained by diagonalization of the covariance matrix, which is not sequential. The analogy is more with a Gramm–Schmidt orthogonalization (Greub, 1975).

Lemma 2. (Transfer of G onto F)

If $T_{G,F} = T_F^{-1}(T_G)$, then $T_{G,F}$ transfers the distribution G onto the distribution F , because $T_G(G) = \mathcal{U}$ and $T_F^{-1}(\mathcal{U}) = F$.

2.3 | Application to the projection of the distribution of Y over the future period P_k

The projection of the distribution of Y_t over the future period P_k relies on another assumption H_3 complementing H_2 and which relates the dynamics of X and Y .

At this stage, different hypotheses concerning time invariance can be made.

Hypothesis H_{3-1} . Let T_{G_k,F_k} , $k \geq 1$ be the transformation transferring the distribution G_k of $(X_{k,t})$ $t \in P_k$, restriction of X_t over P_k , to that of $(Y_{k,t})$, restriction of Y over P_k ; then, T_{G_k,F_k} does not depend on k

$$T_{G_k,F_k} = T_{G,F} \text{ for each } k \text{ in } \mathbb{N}$$

Thus,

$$T_F^{-1}(T_G) = T_{F_k}^{-1}(T_{G_k}) \quad \text{and} \quad T_{F_k}^{-1} = T_F^{-1}(T_G(T_{G_k}^{-1}))$$

and the projection over future period P_k can be obtained through

$$\hat{Y}_{k,t} = T_{F_k}^{-1}(T_{G_k}(X_{k,t})) = T_F^{-1}(T_G(X_{k,t}))$$

with $k \in \mathbb{N}$, $t \in P_k$.

This hypothesis means that the link between both series is invariant in time or that it does not depend on trends.

For univariate bias correction, this is the hypothesis leading implicitly to the same estimation as with the techniques linked to empirical quantile matching (Déqué, 2007).

Hypothesis H_{3-2} . The transformation between two periods is the same for the observation and for the climate model.

It is therefore quite different from our hypothesis H_{3-1} , which states that the transformation between X and Y is invariant in time.

$$T_{F,F_k} = T_{G,G_k} \text{ for every } k \text{ in } \mathbb{N}$$

Thus,

$$T_F^{-1}(T_{F_k}) = T_G^{-1}(T_{G_k}) \quad \text{and} \quad \hat{Y}_{k,t} = T_G^{-1}(T_{G_k}(Y_{0,t}))$$

However, if $Y_{k,t}$ in the future is directly computed from the observations recorded over the observation period, then it will keep the observed interannual variability, because only the distribution is corrected. Thus, to avoid this unrealistic behavior, because there is no reason that interannual variability in the future will mimic that of the recent past period, one can rather compute

$$T_{F_k} = T_F(T_G^{-1}(T_{G_k})),$$

and the desired bias-corrected time series over future period P_k can be obtained through

$$\hat{Y}_{k,t} = (T_{F_k})^{-1}(T_{G_k}(X_{k,t})) = (T_{G_k})^{-1}(T_G(T_F^{-1}(T_{G_k}(X_{k,t}))))$$

with $k \in \mathbb{N}$, $t \in P_k$.

For univariate bias correction, this estimation is the same as the one obtained by the CDFt correction for example (Michelangeli et al., 2009).

We use both approaches in the generalization to multivariate bias correction.

H_1 means that the distributions are stationary over long enough periods P_k . H_3 hypotheses mean either that the link between the deformations of the distributions of the modeled time series X_t and of the observed ones Y_t over these periods of stationarity does not vary over time (H_{3-2}) or that the link between the distributions of X_t and Y_t does not depend on time t (H_{3-1}). However, as H_3 only concerns instantaneous distributions, it does not imply the transfer of the entire dynamic from one time series to the other. This could be done partially, for example, if what has been previously described is not applied to F and G but to the distributions of (Y_{t-1}, Y_t) and (X_{t-1}, X_t) , which could be possible according to H_2 . Nevertheless, intuitively, it can be seen that some trajectory properties will bring additional consistency between F_0 and F_k . For example, if F has two components, Y_t^1 and Y_t^2 , such consistency is due to the consideration of the whole temporal dependency, for example, that of Y_t^1 and Y_{t-1}^2 .

This justifies the idea proposed by different authors (Vrac & Friederichs, 2015) of reordering the time series $Y_{k,t}$, $k \in \mathbb{N}, t \in P_k$ with regard to Y_t , $t \in P_0$. Remaining in dimension 2, if $R_t = (r_t^1, r_t^2)$, $t \in P_0$ is the rank vector computed for each component independently, then the transformations to period P_k are the simple permutations σ^1, σ^2 such that

$$\sigma^1(r_t^1, t \in P_k) = (r_t^1, t \in P_0) \quad \text{and} \quad \sigma^2(r_t^2, t \in P_k) = (r_t^2, t \in P_0).$$

However, this implies that the temporal sequencing of the variables over period P_0 is imposed to the future period P_k , which means, in particular, that the interannual variability in the period P_k remains that of period P_0 . This is a very strong assumption, probably too strong because it is physically very unlikely that this will be the case for climate time series. This is not the case for two historical periods, and it can be anticipated that climate change will impact not only the mean but also the variability, and even the whole dynamics, both daily and interannually.

With our proposed approach, we have

$$\begin{aligned} Y_t &= T_F^{-1}(U_t) \quad \text{for } t \in P_0 \\ Y_{k,t} &= T_{F_k}^{-1}(U_{k,t}) \quad \text{for } t \in P_k, \text{ with:} \\ U_{k,t} &= T_{F_k}(Y_t) \quad \text{and} \quad V_{k,t} = T_{G_k}(X_t). \end{aligned}$$

The components of U_t and V_t are independent, which allows reordering and applying the reordering transformation independently for each component. Such a reordering could be added, but there is still a strong risk of overfitting. This will not be considered here but rather left for a forthcoming paper.

2.4 | Validation

Once the estimations have been made using the functional estimations described in Section 2, validation is undertaken in the following way.

The historical period P_0 is divided into two subperiods Q_0 and Q_1 . Then, Q_0 is used to calibrate F , and G and Q_1 is used for validation. If $\hat{Y}_t$, $t \in Q_1$ is the corrected time series, the aim is to validate the bias correction using both hypotheses H_{3-1} and H_{3-2} , then either

$$\hat{F}_1^{-1} = T_{F_0}^{-1}(T_{G_0}(G_1^{-1}))$$

or

$$\hat{F}_1 = T_{F_0}(T_{G_0}^{-1}(G_1)),$$

F_0 and G_0 being the distribution functions over Q_0 and G_1 over Q_1 .

Validation consists then in comparing $\hat{F}_1$ to F_1 , which can be estimated here. If d is a distance between distributions in $\mathbb{R}^p$, the level of correction will be defined as

$$r = \frac{d(F_1, \hat{F}_1)}{d(F_1, G_1)}.$$

Then, the choice of d may be quite arbitrary. If $H(z) = P(Z^1 \leq z^1, \dots, Z^p \leq z^p)$ is the distribution associated to the distribution H of Z on $\mathbb{R}^p$, the distance that naturally generalizes the distance on $\mathbb{R}$ for continuous distributions is given by $d(H, K) = \sup_{z \in \mathbb{R}^p} |H(z) - K(z)|$. Under the assumption that H and K have densities φ and ψ , a distance L^p like

$$d_p(H, K) = \left(\int_{\mathbb{R}^p} |\varphi(z) - \psi(z)|^p \right)^{1/p}$$

can be used.

These distances have been chosen here, but others could have been used, such as the “divergences” proposed by Rust, Vrac, Lengaigne, and Sultan (2010).

3 | DISTRIBUTION ESTIMATIONS

The distributions F , G , and G_k are at first unknown and still to be estimated, so that the transformations T_F , T_G , and T_{Gk} can be obtained explicitly. This implies the estimation of one-dimensional conditional distributions, but with a conditioning on 2 to $p - 1$ dimensions. According to hypotheses H_1 and H_2 , all considered distributions have a probability density function on $\mathbb{R}^p$, which allows the use of nonparametric smoothing methods such as kernel density estimation techniques. Two main approaches exist to estimate conditional distributions:

1. direct use of a kernel estimator adjusted through an indicator function (Hall, Wolff, & Yao, 1999);
2. estimation of a conditional density and integration afterwards (Hyndmann, Bashtannyk, & Grunwald, 1996).

We will use this last approach, which allows the computation of different validation criteria. The numerical examples in parts 4 and 5 are given for two variables ($p = 2$). Even if the theory is general and valid regardless of the number of dimensions, in practice, the number of values necessary for a reliable estimation of the densities increases in a polynomial way with dimension p .

Let us consider a time series $X_t = (X_t^1, X_t^2)$, which verifies hypotheses H_1 and H_2 . The estimation of the density of X_t^1 is classical and has good asymptotical properties (for a sufficiently large number of data) thanks to hypothesis H_1 . This density will be denoted by $g_1(x^1)$ and its estimator $\hat{g}_1(x^1)$.

Then, $g_{2|1}(x^1, x^2) = \frac{g(x^1, x^2)}{g_1(x^1)}$. The classical kernel density estimator is given by

$$\hat{g}_{2|1}(x^1, x^2) = \frac{\frac{1}{nh^{(1)}h^{(2)}} \sum_{i=1}^n K^{(1)}\left(\frac{x^1 - X_i^1}{h^{(1)}}\right) K^{(2)}\left(\frac{x^2 - X_i^2}{h^{(2)}}\right)}{\frac{1}{nh^{(1)}} \sum_{i=1}^n K^{(1)}\left(\frac{x^1 - X_i^1}{h^{(1)}}\right)},$$

where the kernels $K^{(1)}$ and $K^{(2)}$ are positive functions $\mathbb{R} \rightarrow \mathbb{R}$, with integral 1 and whose square can be integrated. The smoothing parameters $h^{(1)}$ and $h^{(2)}$ are chosen according to the data used for the estimation.

Different types of kernels are generally used:

1. Gaussian: $K(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$;
2. Epanechnikov: $K(x) = \frac{3}{4} (1 - x^2) 1_{\{|x| \leq 1\}}$;
3. Student: $\frac{1}{\sqrt{\pi}} \frac{\Gamma((v+1)/2)}{\Gamma(v/2)} \left(1 + \frac{x^2}{v}\right)^{-\frac{v+1}{2}}$.

We have chosen the Gaussian kernel.

To estimate parameters $h^{(1)}$ and $h^{(2)}$, we use the R package “hdcrcde” developed by Rob Hyndmann and on the basis of the methods described in Hyndmann et al. (1996), Bashtannyk and Hyndmann (2001), Hall et al. (1999), Fan, Yao, and Tong (1996), De Gooijer and Gannoun (2000), Liebscher (1996), and Hyndmann and Yao (2002), valid under hypotheses H_1 and H_2 . The choice of $h^{(1)}$, $h^{(2)}$ is then made either by using asymptotical results based on H_2 , to which a condition C^2 (stating that all partial derivatives of g have to be twice continuously differentiable) has to be added, or by cross-validation using blocks in order to deal with the weak dependence. The criterion to be minimized in order to choose $h^{(1)}$ and $h^{(2)}$ is generally the L^2 norm between the densities and their estimates. We have also used methods based on subsampling and regression (Hall et al., 1999) or kernels with polynomial weights (Bashtannyk and Hyndmann, 2001), and in practice, the approach used in the R package “hdcrcde” (which mixes regression and subsampling for kernel estimation, e.g., Bashtannyk and Hyndmann, 2001; Hyndmann et al., 1996; Hyndmann & Yao, 2002) is efficient, that is, easy to handle and giving good results in a reasonable computing time.

4 | WHEN IS IT JUSTIFIED?

The previously described methodology aims at bias-correcting the joint distribution of two or more variables at the same time. This brings more complexity and needs more data than the usual univariate distribution correction. Thus, in order to judge if there are best suited conditions for the use of such a technique, it has been decided to first test the approach with statistically simulated data.

4.1 | Design of the simulation study

Since shuffling is not considered here, the corrections are only based on the distribution of the data. So, it has been decided to test the approach with data generated by chosen parametric distributions.

The aim is thus to simulate four bivariate distributions:

1. corresponding to the observations over the current period: F_0 ;
2. corresponding to the model simulation over the same current period: G_0 ;
3. corresponding to the model simulation over a future period: G_k ;
4. corresponding to the observations over a future period, which is not known a priori and only used to validate the approach: F_k .

It has been decided to simulate bivariate Gaussian distributions, which is a simple design.

4.2 | Design of the tests

The parameters for the bivariate Gaussian distribution of the current period observations F_0 have been chosen on the basis of summer temperature distributions for two distant points in Europe, chosen arbitrarily as Hamburg and Orly. The means, variances, and covariances are estimated from the E-OBS daily mean temperature time series over the period 1979–2014 (Haylock et al., 2008) for the month of July as follows:

$$m_1 = 17.5, \quad m_2 = 20.0, \quad v_{11} = 10, \quad v_{22} = 9, \quad v_{12} = v_{21} = 6.$$

with m_1 and m_2 respectively the means for variables 1 and 2, v_{11} and v_{22} their variances, and v_{12} their covariance. This leads to a linear correlation $\rho = 0.63$ between both variables.

The values corresponding to the observations over the current period are thus simulated by a bivariate Gaussian distribution with the previously mentioned parameters, and 2,000 values are produced.

Then, some hypotheses have to be made to simulate the data for the model simulations (current and future periods) and for the observations over a future period used for verification.

To do so, model errors have first been postulated, additive for the means and multiplicative for the variances and covariances, denoted by em_1 and em_2 for the mean errors and es_{11} , es_{22} , and es_{12} for the variance and covariance errors. The data corresponding to the model simulations for each variable over the current period (2,000 values) are thus produced by a bivariate Gaussian distribution G_0 with parameters

$$m_1 + em_1, \quad m_2 + em_2, \quad v_{11} \times es_{11}, \quad v_{22} \times es_{22}, \quad v_{12} \times es_{12} (= v_{21} \times es_{21}).$$

Then, climate shifts due to climate change are postulated in the same way, that is, as additive for the means and multiplicative for the variances and covariances, and denoted by dm_1 , dm_2 , ds_{11} , ds_{22} , and ds_{12} , respectively. The 2,000 values corresponding to the model simulation over a future period are produced by a bivariate Gaussian distribution G_k with parameters

$$m_1 + em_1 + dm_1, \quad m_2 + em_2 + dm_2, \quad v_{11} \times es_{11} \times ds_{11}, \quad v_{22} \times es_{22} \times ds_{22}, \quad v_{12} \times es_{12} \times ds_{12},$$

and the data corresponding to the observations over a future period (2,000 values) are produced by a bivariate Gaussian distribution F_k with parameters

$$m_1 + dm_1, \quad m_2 + dm_2, \quad v_{11} \times ds_{11}, \quad v_{22} \times ds_{22}, \quad v_{12} \times ds_{12} (= v_{21} \times ds_{21}).$$

The validation criterion is the ratio r defined in section 2.4 with both L_1 and L_∞ norms, which is the distance between bias-corrected and observed distributions divided by the distance between simulated and observed distributions, both estimated over the future period.

The aim of the applied bias corrections, either univariate or bivariate, is to estimate the bivariate distribution of the observations for a future period ($\hat{F}_k$) from that of the observations and model simulation over the current period (F_0 and G_0) and of the model simulation over a future period (G_k). Then, the previously defined ratio r measures the performance of the correction in making the corrected distribution closer to that of the observations over future period F_k than the distribution of the model simulation over future period G_k if $r < 1$.

Six bias corrections are applied and compared, denoted by UH, UV, B12H, B21H, B12V, and B21V, respectively.

1. Each variable is independently corrected under hypothesis H_{3-1} (stationarity of the transformation between model and observations).
2. Each variable is independently corrected under hypothesis H_{3-2} (stationarity of the transformation between present and future periods).
3. Bivariate correction under hypothesis H_{3-1} with variable 1 corrected first.
4. Bivariate correction under hypothesis H_{3-1} with variable 2 corrected first.
5. Bivariate correction under hypothesis H_{3-2} with variable 1 corrected first.
6. Bivariate correction under hypothesis H_{3-2} with variable 2 corrected first.

UH and UV are similar to empirical quantile mapping and CDFt, respectively. Here, because we only deal with distributions, the bivariate distribution correction under hypothesis H_{3-2} is computed directly from Y_0 through $T_G^{-1}(T_{G_k})$. The univariate bias corrections are computed in the same way as the correction of the first variable in the bivariate corrections, and not taken from the R packages for quantile mapping or CDFt, in order to remain consistent in the comparisons.

Two cases have been considered in order to better discriminate hypotheses H_{3-1} and H_{3-2} and the consequences of using a correction technique, which may not be the best adapted. As a matter of fact, when dealing with climate model simulations, it is difficult to test which stationarity is best verified (because we do not have the observations in the future), and quantile mapping or CDFt is generally indifferently used. The chosen cases correspond to the following:

1. one case with model errors larger than climate change;
2. one case with climate change larger than model errors.

In order to be able to test the order of variable corrections in the bivariate corrections, the errors and climate change shifts are not equal for each variable. The test cases are made with the following parameters:

1. $em_1=4$, $em_2=5$, $es_{11}=es_{22}=es_{12}=0.5$ and $dm_1=2$, $dm_2=1.5$, $ds_{11}=ds_{22}=1.2$, $ds_{12}=1$;
2. $em_1=1$, $em_2=1.5$, $es_{11}=0.8$, $es_{22}=0.75$, $es_{12}=0.9$ and $dm_1=4$, $dm_2=5$, $ds_{11}=ds_{22}=1.5$, $ds_{12}=1$.

The choices have been made in considering the observed model errors and climate change shifts. Climate models tend to underestimate the variances, and in summer, both the means and the variances increase. Then, this behavior has been exaggerated to produce large errors and large climate shifts.

4.3 | Results

As previously described, we first simulate 2,000 values with Gaussian bivariate distributions using fixed means and variances/covariances, from which we then nonparametrically estimate bivariate distributions to compute distances and compare the different bias corrections. The nonparametrical estimation is based on the R function “kde2d” of package MASS, and with 2,000 values only, such an estimation is uncertain. It has thus first been verified that the distances between the bivariate distributions with the chosen errors and climate change shifts are significantly larger than the distances between two sets of 2,000 points taken from the same distribution. This is preferred to the consideration of a much larger number of values, because first, this considerably increases the computing time for the corrections, and second, in climate change studies, when corrections have to be made, generally done on a monthly basis, we do not dispose of much more values. Thus, four sets of 2,000 points are produced by use of a bivariate Gaussian distribution: one mimicking two variables as observed under current climate conditions, another for current climate as simulated by a climate model, and the two other sets mimicking observed and modeled variables under future climate conditions. Then, the previously defined six bias corrections are applied to the set corresponding to the modeled variables under future climate conditions, so that its distribution gets closer to that of the variables corresponding to the observed ones under future climate conditions (as we are here in an academic situation where all four distributions are simulated). This is done 30 times, and the distributions of the r ratios are examined.

4.3.1 | Errors larger than climate change

In order to test for the best suited stationarity hypothesis, the transformations based on hypotheses H_{3-1} and H_{3-2} are applied to the same sample. H_{3-1} states that $T_F^{-1}(T_G) = T_{F_k}^{-1}(T_{G_k})$; thus, both $T_F^{-1}(T_G)$ and $T_{F_k}^{-1}(T_{G_k})$ are applied to the sample obtained with G_0 . Then, the distance between the obtained distributions is computed (as L_1 or L_∞ norm). Similarly, H_{3-2} postulates that $T_F^{-1}(T_{F_k}) = T_G^{-1}(T_{G_k})$; thus, both $T_F^{-1}(T_{F_k})$ and $T_G^{-1}(T_{G_k})$ are applied to the sample obtained with F_0 , and the distance between

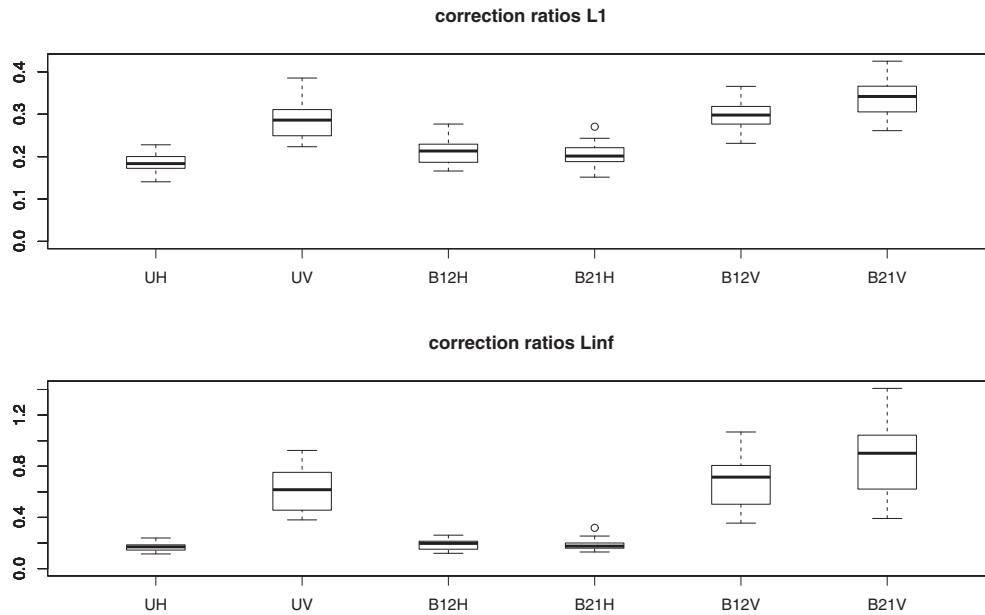


FIGURE 1 Boxplots of the correction ratios obtained with 30 simulations with bivariate Gaussian distributions in the case of larger model errors than climate shifts, based on distance L_1 (top panel) and L_∞ (bottom panel). U refers to univariate correction and B to bivariate starting with variable 1 (12) or 2 (21), and H is for hypothesis H_{3-1} and V for hypothesis H_{3-2}

the obtained distributions is computed. Each transformation is computed with variable 1 first and with variable 2 first, and 30 generations are considered. Then, the distance corresponding to H_{3-1} is compared to that corresponding to H_{3-2} to infer the best verified hypothesis. In this case, the average L_1 distances corresponding to each transformation for the 30 tests are as follows:

distance according to H_{3-1} with variable 1 first: $6.2 \cdot 10^{-4}$,
 distance according to H_{3-1} with variable 2 first: $6.3 \cdot 10^{-4}$,
 distance according to H_{3-2} with variable 1 first: $7.5 \cdot 10^{-4}$,
 distance according to H_{3-2} with variable 2 first: $8.4 \cdot 10^{-4}$.

The distances after H_{3-1} are lower, so hypothesis H_{3-1} seems best verified. Each of the six corrections is then applied to estimate F_k , and Figure 1 presents the boxplots of the r ratio for each correction using either L_1 (top panel) or L_∞ (bottom panel) distances. It shows that, as expected, both univariate and bivariate corrections based on hypothesis H_{3-1} (i.e., UH, B12H, and B21H) perform better than those based on hypothesis H_{3-2} (i.e., UV, B12V, and B21V), although these last corrections bring some improvement too. In this case, the correction is very efficient (mean ratios around 0.2), and the bivariate correction does not bring substantial improvement.

4.3.2 | Climate change larger than model errors

In this case, the mean distances between the different transformations are as follows:

distance according to H_{3-1} with variable 1 first: $6.5 \cdot 10^{-4}$,
 distance according to H_{3-1} with variable 2 first: $8.2 \cdot 10^{-4}$,
 distance according to H_{3-2} with variable 1 first: $2.9 \cdot 10^{-4}$,
 distance according to H_{3-2} with variable 2 first: $3.6 \cdot 10^{-4}$.

Hypothesis H_{3-2} is here best verified. Again, the obtained results after applying each of the six distribution corrections (Figure 2) confirm that the corrections based on hypothesis H_{3-2} perform better in this case. The corrections are however much less efficient than in the previous setting, with best ratios around 0.5, whereas they were around 0.2 in the previous case, but this may be due to the chosen parameters. Actually, the choices made for the model errors and climate change shifts are not symmetrical. Here, choosing the least verified hypothesis may lead to ratios higher than 1 (no corrections). Here, bivariate correction, at least when starting with variable 1, brings some improvement compared to independent univariate corrections.

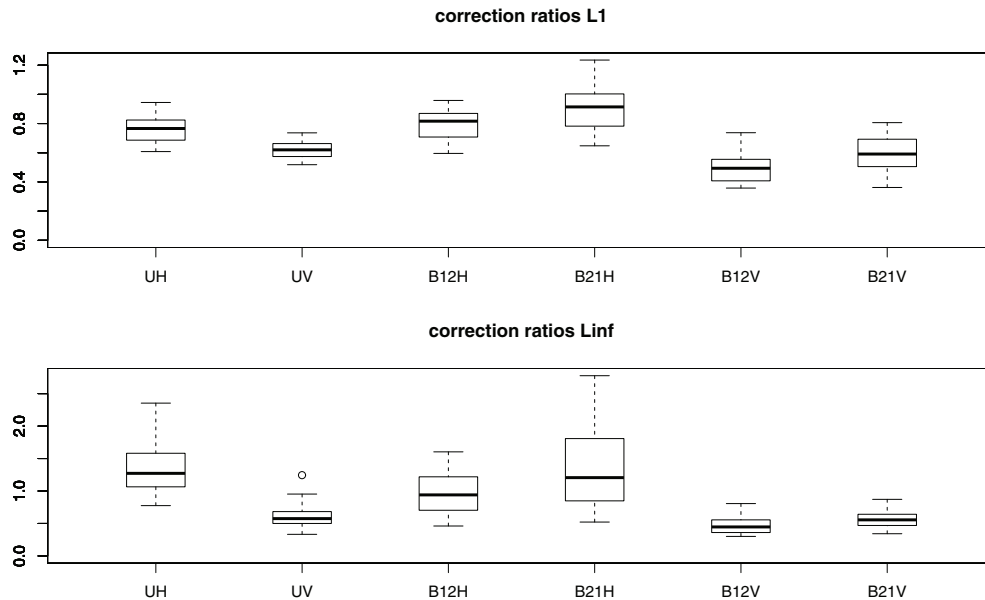


FIGURE 2 Boxplots of the correction ratios obtained with 30 simulations with bivariate Gaussian distributions in the case of larger climate shifts than model errors, based on distance L_1 (top panel) and L_∞ (bottom panel). U refers to univariate correction and B to bivariate starting with variable 1 (12) or 2 (21), and H is for hypothesis H_{3-1} and V for hypothesis H_{3-2}

In both cases, the improvements due to the most suited correction appear more clearly with distance L_∞ . Distance L_1 is an average over the whole distribution (or at least its estimation on some grid), whereas distance L_∞ corresponds to a single value: the maximal one. As here, the introduced errors imply distribution shifts, the corrections reduce the shift and thus have a greater impact on L_∞ than on L_1 .

4.3.3 | The role of correlation strength

Another question arising in this context is the role of the correlation strength between the variables in the importance and performance of multivariate bias correction. In order to investigate this point, the previous tests have been performed again with lower covariances between both variables as follows:

$$\begin{aligned} v_{12}=v_{21}=3, & \text{ which leads to a linear correlation coefficient } \rho \text{ around } 0.3, \text{ and} \\ v_{12}=v_{21}=1, & \text{ which leads to a linear correlation coefficient } \rho \text{ around } 0.1, \end{aligned}$$

respectively called medium and low correlation, whereas the previous test is called high correlation. The results are presented for L_1 and L_∞ norms in Figure 3, in keeping only the best corrections in each case (H for hypothesis H_{3-1} in the case of larger model errors than climate shifts and V for hypothesis H_{3-2} in the reverse case). They show that the correlation strength does not have any significant impact on the performance of the corrections, even though, in the case of climate shifts larger than model errors, bivariate correction performs better than univariate correction in the strong correlation case but equally well otherwise. Correlation has been compared too (Figure 4). Whereas univariate correction does not really change the model correlation, bivariate correction does, and generally in the right way. However, bivariate correction seems to underestimate the correlation coefficient when covariance (and correlation) is high, especially when hypothesis H_{3-2} is concerned. Moreover, bivariate correction shows more variable results than univariate correction among the 30 bivariate Gaussian distribution generations. This is most probably due to the fact that bivariate correction needs more distribution estimations than univariate correction, which increases statistical errors.

Now, when both model errors and climate shifts are moderate, that is,

$$em_1 = 1, \quad em_2 = 1.5, \quad es_{11} = 0.8, \quad es_{22} = 0.75, \quad es_{12} = 0.9, \quad \text{and} \quad dm_1 = 2, \quad dm_2 = 1.5, \quad ds_{11} = ds_{22} = 1.2, \quad ds_{12} = 1$$

then both corrections perform as well, as can be seen in Figure 5, because both hypotheses are quite equally verified. However, the dispersion for the bivariate correction is higher again, which may be linked to the uncertainty in the bivariate distribution estimation when the correlation is high. As a matter of fact, with these parameters, the correlation between both variables is

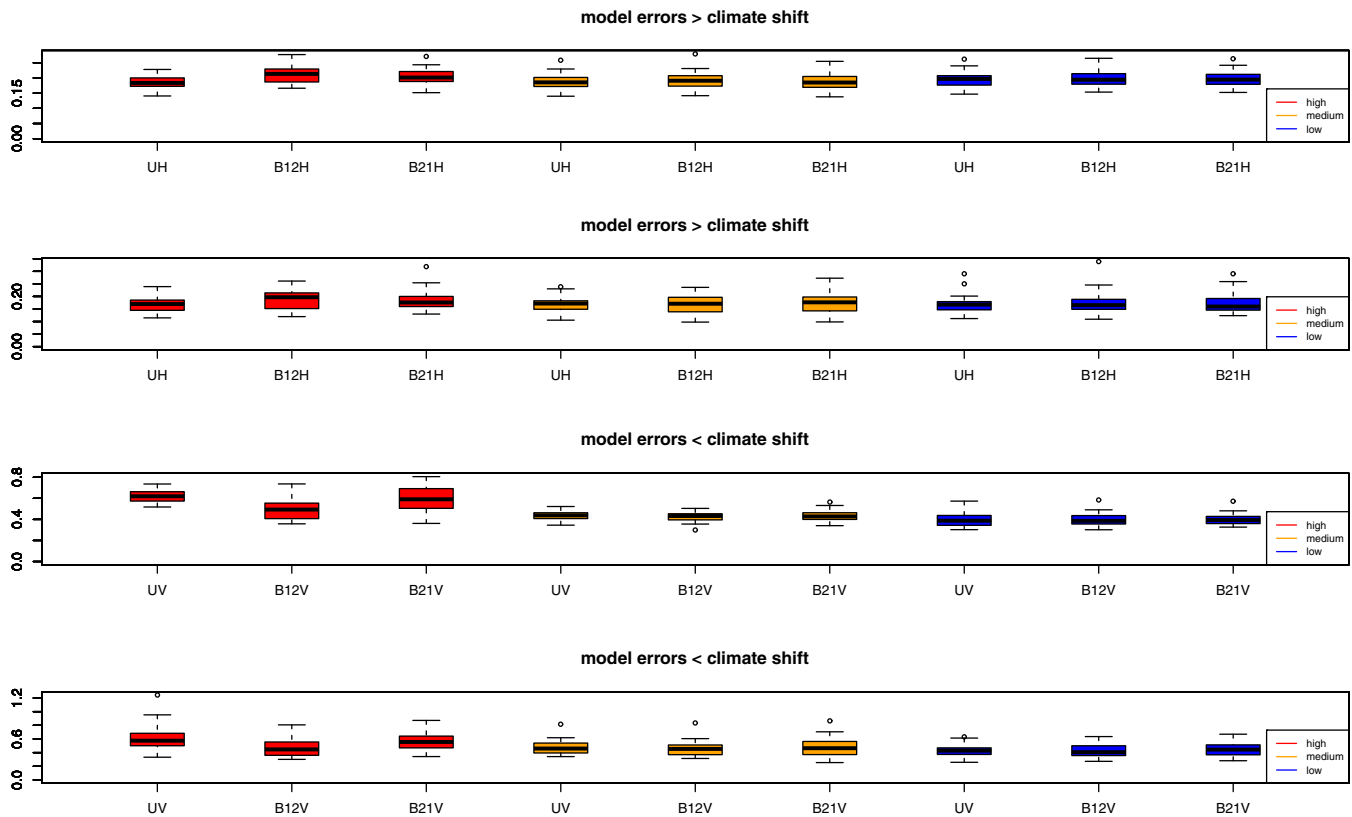


FIGURE 3 Boxplots of the correction ratios obtained with 30 simulations with bivariate Gaussian distributions in both cases (top panel: larger model errors than climate shifts; bottom panel: larger climate shifts than model errors) based on distances L_1 and L_∞ and for different correlation strengths between the variables: high (red), medium (orange), and low (blue). U refers to univariate correction and B to bivariate starting with variable 1 (12) or 2 (21), and H is for hypothesis H_{3-1} and V for hypothesis H_{3-2}

0.63 for F_0 and goes to 0.73 for G_0 . This was the same for the case with higher climate shifts than model errors, which showed a similar behavior (Figure 2).

5 | TEST WITH CLIMATE DATA

The goal of the developed approach is the bias correction of climate data; thus, the next step consists in testing it with observed and model variables. As all here is defined for continuous distributions, and in order to test the approach with different distributions for each variable, it has been decided to work with temperature and wind speed. Long time series of observed daily mean temperature and wind speed have been obtained for the city of Hamburg in Germany from the European Climate Assessment and Dataset (ECA&D) project website (<http://eca.knmi.nl/dailydata/>; Klein Tank et al., 2002). In order to maximize the chance that the time series are homogeneous, they will be considered from 1950 to 2015 (66 years). Then, the climate model simulation of the IPSL-CM5-MR model (Dufresne et al., 2013) has been arbitrarily chosen in the Coupled Model Intercomparison Project 5 (CMIP5) database as a test model, and the time series of the nearest grid box to Hamburg has been extracted from the historical run (1950–2005) and the Representative Concentration Pathway (RCP) 8.5 projection run (2006–2100), to compute time series of daily mean temperature and wind speed over the same period $P_0=1950-2015$ (see Supporting information). This period is then divided into two 33-year subperiods: $Q_1=1950-1982$, chosen as the calibration period, and $Q_2=1983-2015$, chosen as the validation period.

As the approach is valid for stationary time series over the defined periods, the correction is applied on a monthly basis, in order to get rid of the annual cycle. It is always difficult to consider that climatic variables are stationary over a defined period, because of both climate change and interannual variability. The World Meteorological Organization recommends considering at least 30 years to define the climate of an area; thus, considering 33-year periods seems a reasonable choice.

First, the distances after transformations of the same sample according to both H_{3-1} and H_{3-2} hypotheses are computed for each month to infer which one is best verified. Table 1 summarizes the results for each month, together with an indication of the correlation between wind and temperature, as observed over period Q_1 , according to the linear Pearson correlation coefficient or

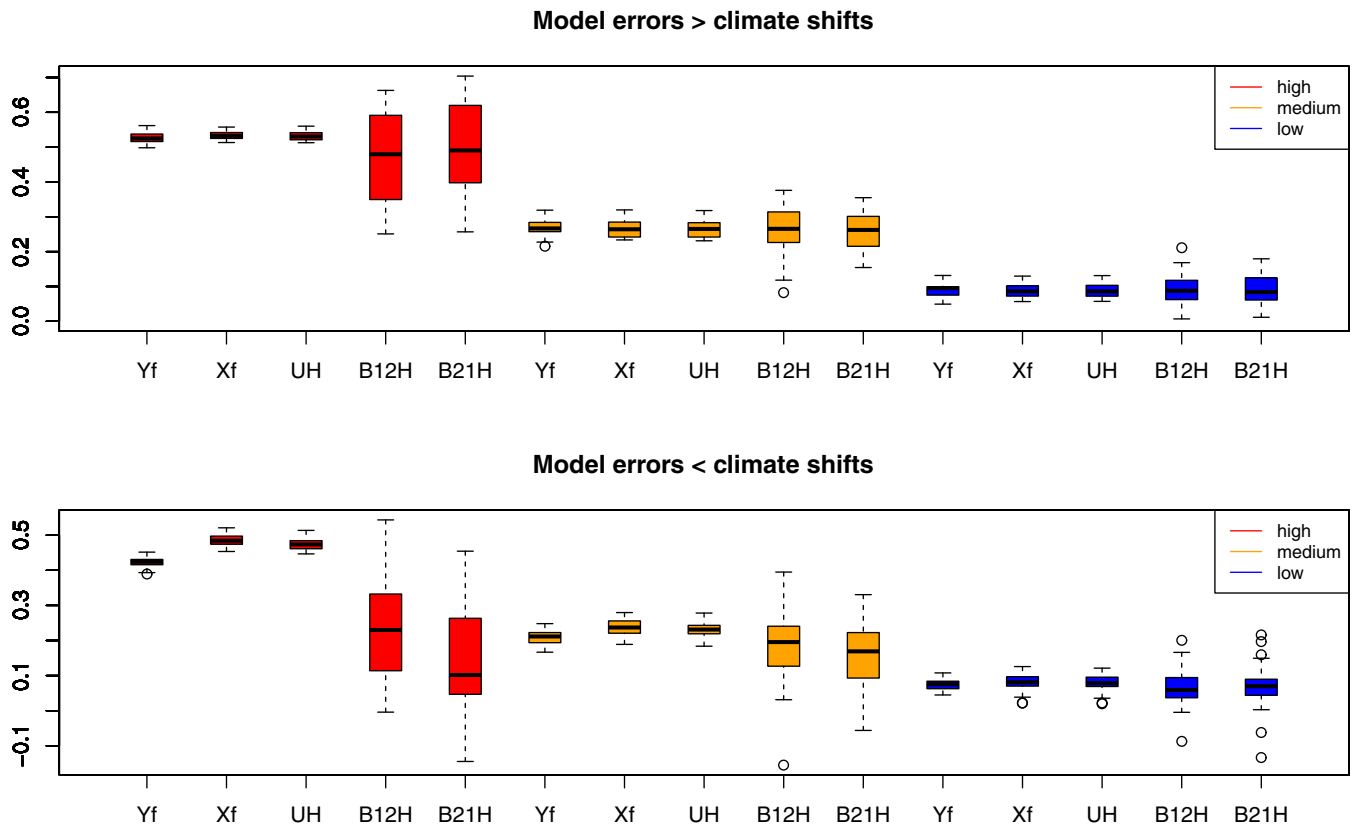


FIGURE 4 Boxplots of the correlation coefficients obtained with 30 simulations with bivariate Gaussian distributions in both cases (top panel: larger model errors than climate shifts; bottom panel: larger climate shifts than model errors) and for different covariance strengths between the variables: high (red), medium (orange), and low (blue). Yf refers to the observations over the validation period and Xf to the simulation over the same period, U refers to univariate correction and B to bivariate starting with variable 1 (12) or 2 (21), and H is for hypothesis H_{3-1} and V for hypothesis H_{3-2}

the rank Spearman one. Except for July, for which H_{3-1} seems clearly best verified, for all other months, either H_{3-1} shows a small preference or it is difficult to discriminate both hypotheses. Therefore, the multivariate correction will be applied according to H_{3-1} , which corresponds to estimations similar as those made by empirical quantile mapping, because between the two recent past periods, the climate shift is not too high. Here, the functions of the R package *qmap* have been used for the univariate corrections in order to compare our proposed approach to standard ones used in climate studies. Daily temperature and wind speed are generally moderately correlated (positively in winter, negatively in summer) except in September.

5.1 | Temperature first, then wind

The first test is made by correcting temperature first and then wind according to temperature, as described in Sections 1 and 2. The ratios of the distance between the bivariate distribution of the corrected variables and the observed ones divided by the distance between the modeled variables and the observed ones over the validation period are compared for both bivariate and independent univariate bias corrections for each month (Table 2). Here, only the L_1 norm is considered as both used norms lead to the same conclusions in the previous section. The first outcome is that, generally, bias correction improves the distance to the observations (ratio < 1), and bivariate correction gives slightly better results than univariate corrections for seven months. The worst correction is obtained for the month of May, whereas the best occurs in July, both for univariate and bivariate corrections. Figure 6 allows the comparison of the bivariate distributions in May, for the observations and model over the calibration (Q_1) and validation (Q_2) periods and for Q_1 and Q_2 for the model and observations. It shows that the model errors are rather low and that there is very little change between both periods, for the model as well as for the observations. It is thus not surprising that the correction is modest in this case, as shown in Figure 7. On the contrary, in July (Figure 8) the model errors are quite large, and climate change is modest. The situation is more similar to that of our academic case with large model errors and moderate climate shift, which previously lead to the best corrections based on hypothesis H_{3-1} (Section 4.3.1). Figure 9 illustrates the distributions before and after corrections and shows that bivariate correction brings some improvement. As far as correlation

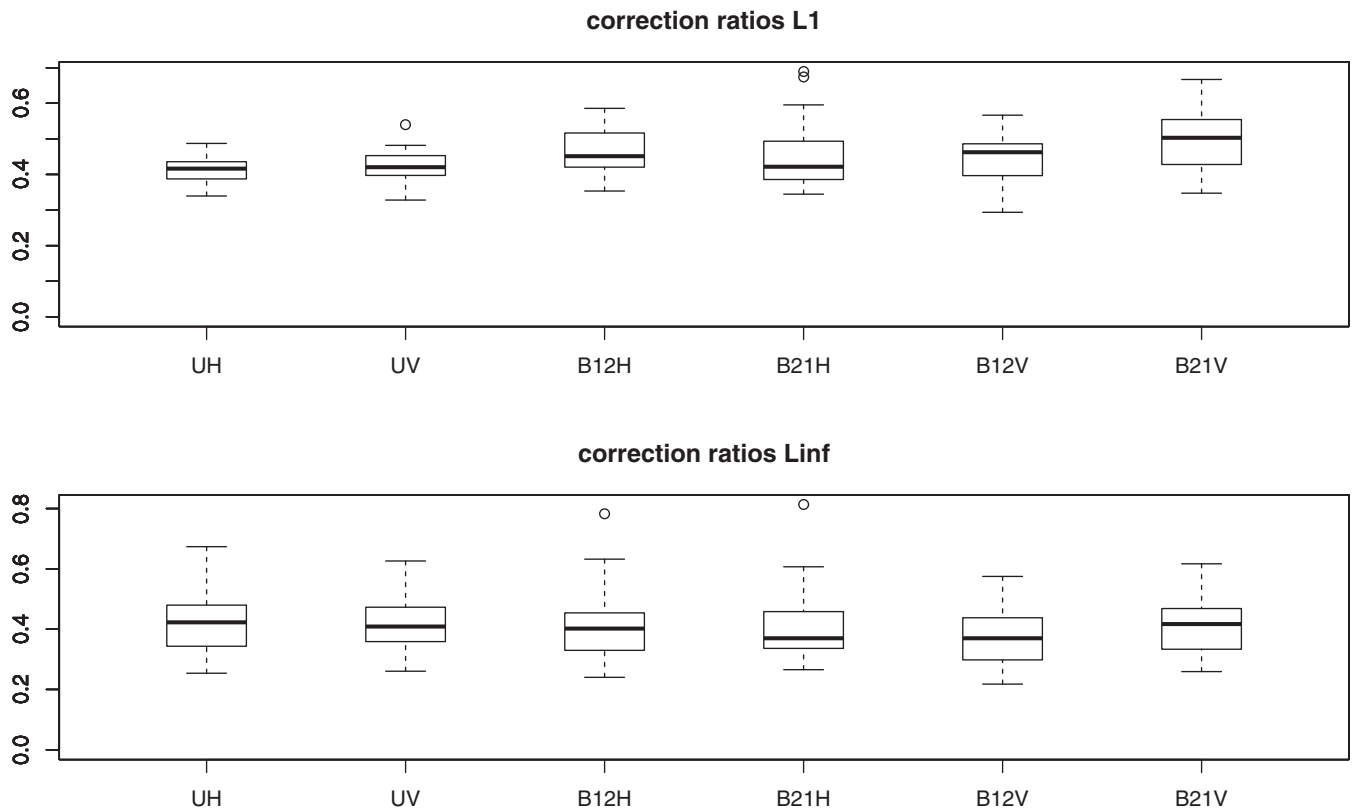


FIGURE 5 Boxplots of the correction ratios obtained with 30 simulations with bivariate Gaussian distributions in the case of moderate climate shifts and model errors, based on distance L_1 (top panel) and L_∞ (bottom panel). U refers to univariate correction and B to bivariate starting with variable 1 (12) or 2 (21), and H is for hypothesis H_{3-1} and V for hypothesis H_{3-2}

TABLE 1 L_1 distance between the bivariate distributions obtained with transformations based on hypothesis H_{3-1} with temperature (T) or wind (w) first (first two columns) and between the bivariate distributions obtained with transformations based on hypothesis H_{3-2} with temperature (T) or wind (w) first (columns 3 and 4) and the correlation between temperature and wind (Pearson and Spearman coefficients for the observations (columns 6 and 7) and for the IPSL model simulation (columns 8 and 9) for each month

Month	H_{3-1} T first	H_{3-1} w first	H_{3-2} T first	H_{3-2} w first	Correlation			
	$L_1 (10^{-4})$	$L_1 (10^{-4})$	$L_1 (10^{-4})$	$L_1 (10^{-4})$	Observations		IPSL simulation	
					Pearson	Spearman	Pearson	Spearman
January	7.0	6.6	9.5	9.0	0.47	0.50	0.50	0.54
February	7.0	6.3	10.5	8.1	0.36	0.35	0.40	0.43
March	14.7	13.2	14.0	13.8	0.11	0.10	0.17	0.18
April	18.0	17.1	17.1	17.3	-0.15	-0.15	-0.07	-0.06
May	13.7	12.6	19.0	17.6	-0.17	-0.17	-0.32	-0.33
June	21.5	20.8	21.6	21.3	-0.28	-0.28	-0.27	-0.26
July	13.2	12.2	26.2	28.5	-0.37	-0.39	-0.19	-0.15
August	15.0	16.2	17.3	20.0	-0.21	-0.22	-0.32	-0.30
September	14.5	14.8	19.3	22.9	-0.01	0.00	-0.14	-0.13
October	13.3	9.6	10.5	9.3	0.17	0.16	0.10	0.11
November	8.2	11.6	12.2	13.3	0.33	0.33	0.39	0.40
December	9.4	8.7	11.7	11.2	0.46	0.48	0.47	0.50

is concerned, univariate bias correction does not have any impact on the model correlation, whereas bivariate correction does, and generally improves the correlation. This can be seen in Table 2, for example, for May, when the model anticorrelation is stronger than observed and better after bivariate correction, or for July, when model anticorrelation is weaker than observed and

TABLE 2 Correlation (Spearman rank correlation coefficient) over the validation period and ratio of the distance (between the bivariate distributions of the corrected and observed variables divided by that of the modeled and observed variables) and correlation over the validation period after bivariate bias correction with temperature corrected first and independent univariate bias correction. Bold indicates the lowest ratios obtained when correction is efficient. Italic means that the correction is not efficient (ratio > 1)

Month	Correlation		Bivariate correction		Univariate correction	
	Observations	IPSL model	Distance ratio	Correlation	Distance ratio	Correlation
January	0.55	0.55	0.522	0.62	0.563	0.55
February	0.44	0.43	0.328	0.39	0.301	0.43
March	0.11	0.18	0.758	0.17	0.771	0.18
April	−0.08	−0.05	0.713	−0.14	0.712	−0.05
May	−0.11	−0.30	0.992	−0.21	<i>1.098</i>	−0.30
June	−0.26	−0.29	0.496	−0.33	0.483	−0.29
July	−0.31	−0.16	0.179	−0.40	0.244	−0.16
August	−0.20	−0.34	0.600	−0.36	0.606	−0.34
September	0.05	−0.12	0.855	−0.03	0.845	−0.12
October	0.15	0.12	0.934	0.17	0.952	0.12
November	0.32	0.36	0.788	0.36	0.849	0.36
December	0.49	0.47	0.671	0.45	0.650	0.47

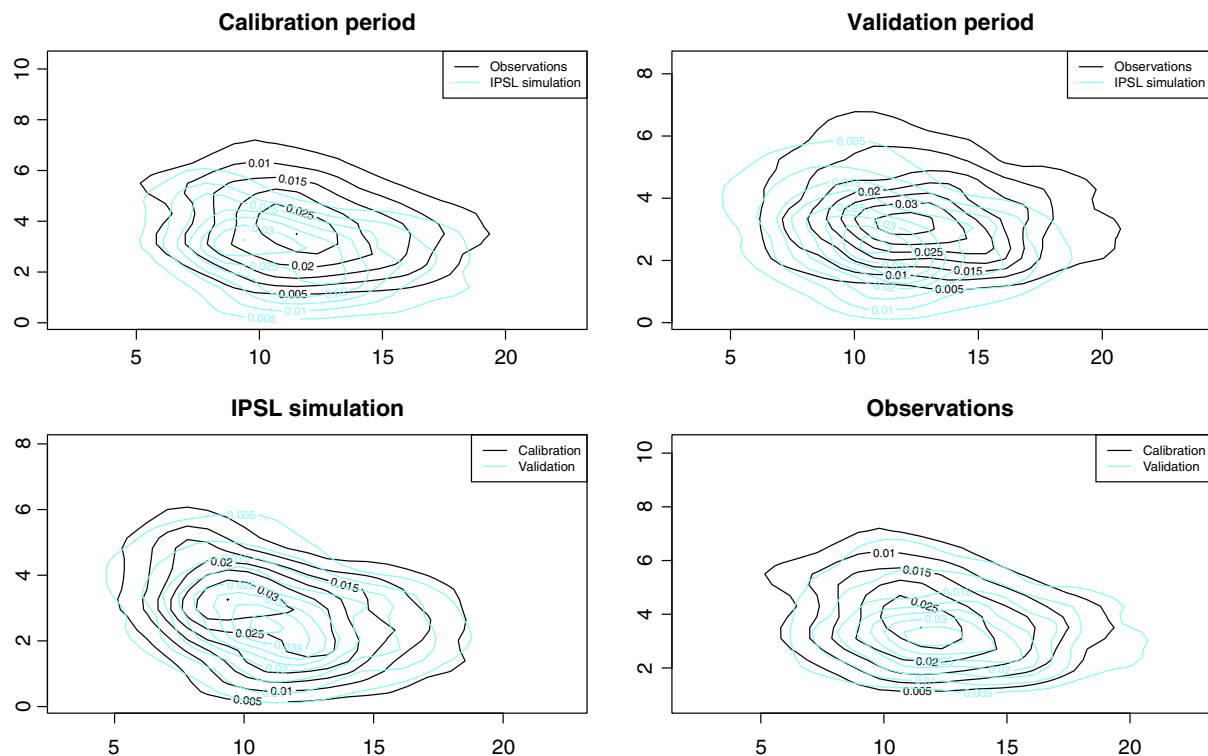
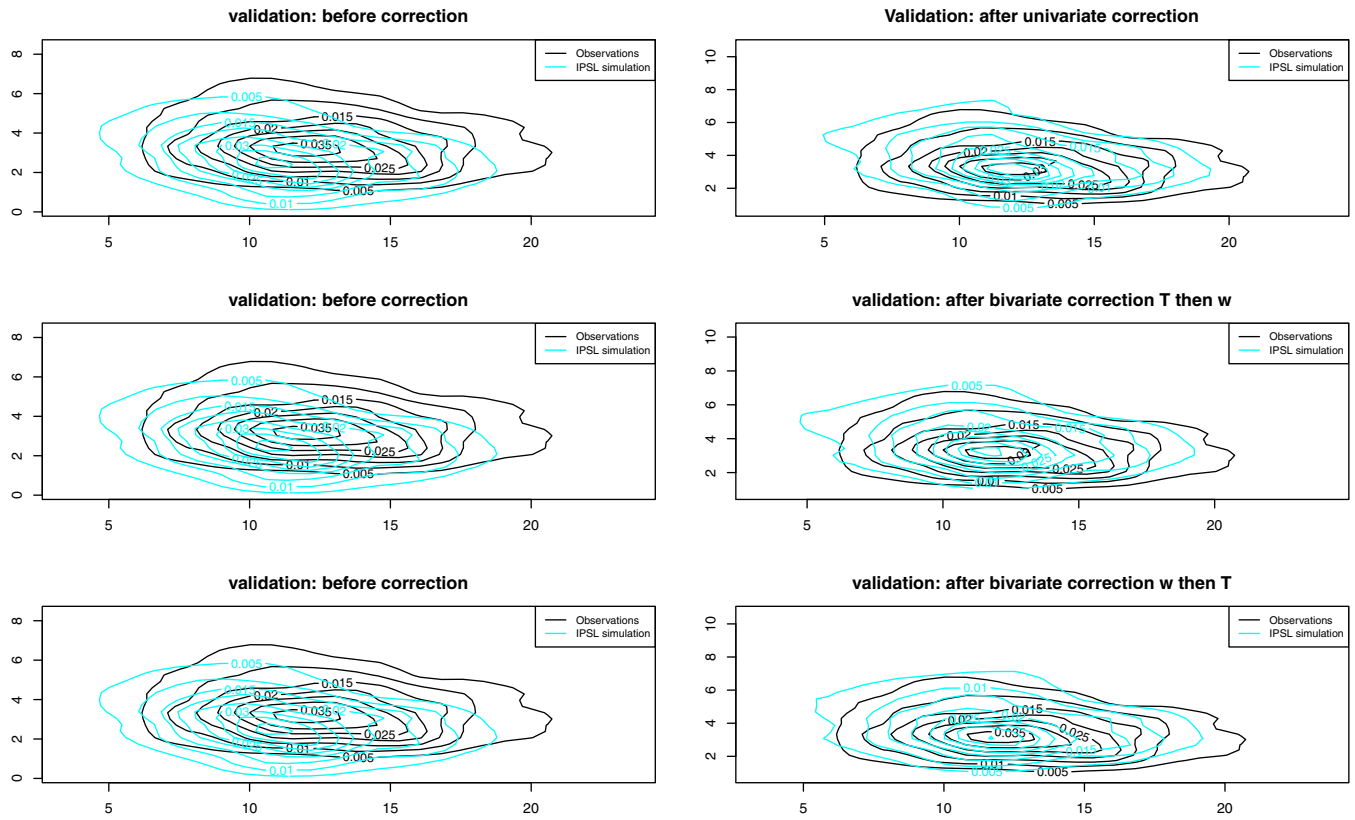


FIGURE 6 Top line: comparison between observed (black) and modeled (cyan) bivariate distributions of temperature and wind speed in Hamburg for each period (calibration period: left panel; validation period: right panel); bottom line: comparison between calibration (black) and validation (cyan) bivariate distributions of temperature and wind speed in Hamburg for the model and the observations (IPSL model: left panel; observations: right panel) for the month of May

increased by the bivariate bias correction. This is shown for the Spearman rank correlation coefficient, but the results are similar with the Pearson correlation coefficient. Thus, bivariate corrections clearly improve the situation if the correlations for the observations and model are different enough to allow the correction to be larger than the statistical errors due to the dimension.



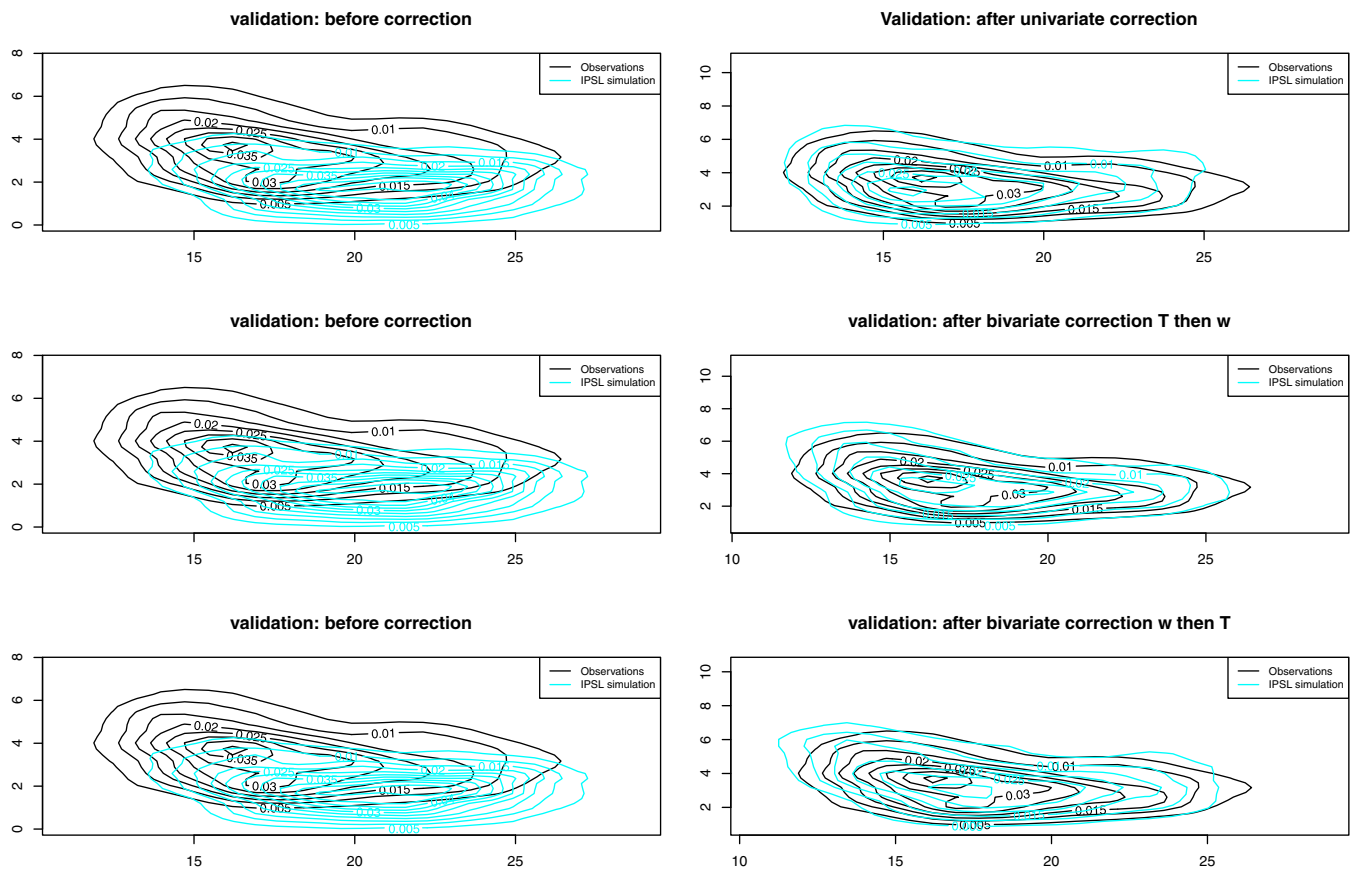


FIGURE 9 Same as Figure 7 but for the month of July

TABLE 3 Same as Table 2 but for wind corrected first, then temperature according to wind

Month	Correlation		Bivariate correction		Univariate correction	
	Observations	IPSL model	Distance ratio	Correlation	Distance ratio	Correlation
January	0.55	0.55	0.532	0.48	0.563	0.55
February	0.44	0.43	0.316	0.33	0.301	0.43
March	0.11	0.18	0.729	0.12	0.771	0.18
April	−0.08	−0.05	0.695	−0.11	0.712	−0.05
May	−0.11	−0.30	0.968	−0.19	1.098	−0.30
June	−0.26	−0.29	0.523	−0.35	0.483	−0.29
July	−0.31	−0.16	0.196	−0.39	0.244	−0.16
August	−0.20	−0.34	0.571	−0.36	0.606	−0.34
September	0.05	−0.12	0.853	−0.09	0.845	−0.12
October	0.15	0.12	0.970	0.19	0.952	0.12
November	0.32	0.36	0.819	0.30	0.849	0.36
December	0.49	0.47	0.601	0.58	0.650	0.47

5.2 | Wind first, then temperature

The same corrections have then been tested again but by correcting wind speed first and then temperature according to wind speed. The ratios of distances after and before correction are summarized in Table 3, together with the Spearman rank correlation coefficients. The results are similar even though the months for which univariate correction gives slightly better results are not always the same. July shows again the best performance for the corrections, whereas May remains the worst corrected, with a small advantage to bivariate correction though.

6 | CONCLUSION AND PERSPECTIVES

In this paper, a new approach for bias-correcting multivariate distributions of climate model simulations according to observations has been proposed and tested in controlled conditions, by use of statistical simulations, and in real conditions, with a climate model simulation and observations for wind speed and temperature. This approach is based on the Lévy–Rosenblatt lemma and generalizes the univariate distribution corrections like empirical quantile mapping or CDFt.

The development of the technique first showed that depending on the hypothesis made (i.e., invariance of the transformation between model and observations in time or invariance of the transformation between two periods for observed and model variables), the correction is similar to empirical quantile mapping or to CDFt.

Then, the tests with bivariate Gaussian distributions allowed comparing the performances of the corrections in controlling the model errors and climate shifts, as well as the correlation strength of the variables. The parameters are based on the observed and model July temperature for two distant points, and willingly exaggerated in order to better see the differences in the approaches. It shows that the verification of the chosen stationary hypothesis, that is, H_{3-1} (stationarity of the link between model and observations) or H_{3-2} (stationarity of the link between present and future conditions), has a stronger importance for the correction performance than univariate or bivariate correction, whatever the correlation strength between both variables. Furthermore, the order of variable corrections does not seem to have important consequences in this framework.

Then, a last test is made in a more real setting, with daily mean temperature and wind speed time series observed over period 1950–2015 and simulated by the IPSL-CM5-MR model in Hamburg. The corrections are applied on a monthly basis, in order to meet as closely as possible the hypothesis of stationarity, in a cross-validation setting, 1950–1982 being the calibration period and 1983–2015 the validation period. Daily temperature and wind speed are moderately correlated in each month, with a positive correlation in winter and a negative one in summer, with lower correlation in fall and spring (and almost no correlation in September). As the periods are close, the conditions of a more reliable application of corrections based on hypothesis H_{3-1} are generally met, and this technique is applied. The results show that bivariate correction generally leads to a slightly better correction.

This study opens some questions before the application of such a correction can be generalized. First, it can be extended to the case of at least one variable with noncontinuous distribution. Most applications have to deal with temperature and rainfall rather than temperature and wind speed. Two ways can be tested to do so:

1. correction of the number of rainy days (based on the differences between model and observations if H_{3-1} is considered or on the change between present and future if H_{3-2} is considered), then of the amount of rainfall on rainy days and finally of temperature according to rainfall (in managing again rainy and non-rainy days);
2. transformation of the rainfall distribution, so that it becomes more continuous, as proposed in Vrac, Noël, and Vautard (2016).

Then, although the methodology is generic and theoretically works regardless of the number of variables, extension to more than two variables will need more data for the estimations to be reliable. Furthermore, the need to estimate more distributions increases the statistical errors, and the improvement is more obvious if the discrepancies to be corrected are large.

Finally, the very important question of stationarity remains. Applying the correction on a monthly basis is the easiest solution. However, because of interannual variability, it is necessary to calibrate the correction over a long enough period (at least 30 years). Then, in the climate change context, the conditions of the application of empirical-quantile-mapping-like approaches (based on hypothesis H_{3-1}), that is, the distributions are invariant in time, cannot hold, and then, CDFt-like approaches (based on hypothesis H_{3-2}) are more adapted, but may bring a lower improvement. Besides, using quantile-mapping-like corrections in such cases may worsen the situation. It seems then important to think of techniques able to stationarize the distributions and let them be closer in time and between the observations and model. This can be done by removing seasonalities and trends, at least in the mean and the variance. However, part of the bias is embedded in the estimation of such quantities for the model time series, and they have to be corrected too. In a univariate context, this can be made in an additive or multiplicative way. However, in a multivariate context, this implies thinking of a way of consistently correcting these parts of the signal as well. Future work will consist in clarifying these questions of nonstationarity by working with a parameterization of the two kinds of dynamic deformations we have formalized in hypotheses H_{3-1} and H_{3-2} . Such parameterization should be more complex than a simple shift but still simple enough to be applied routinely.

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SUPPORTING INFORMATION

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